Trend research — 2026-04-19 · incr 18

majority_trend_bars is gated by the 40% majority floor, not the body-ratio threshold

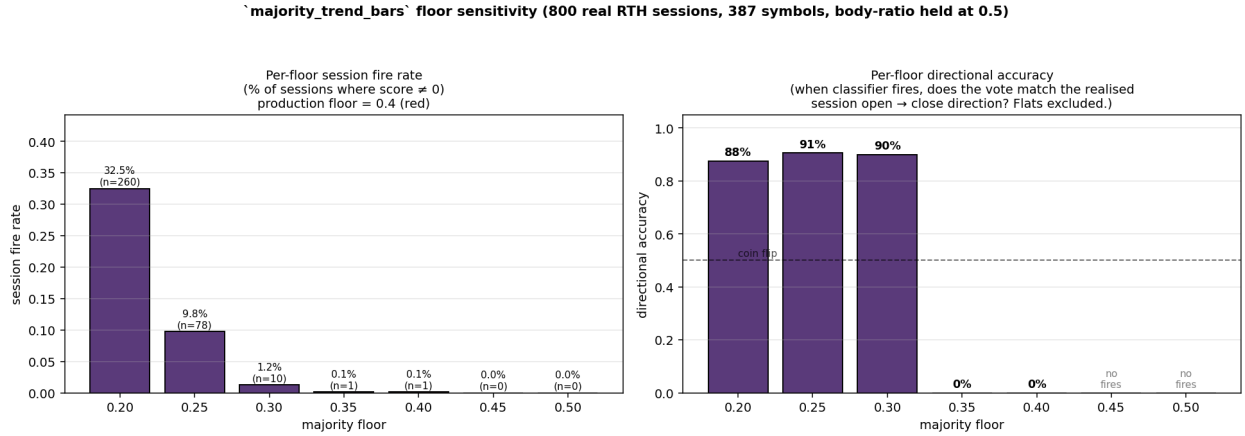
TL;DR

Took the incr-17 finding that **majority_trend_bars** is constant 0 on real data and ran the recalibration study on **800 RTH 5-min equity sessions across 387 symbols**. Two-axis sweep — body-ratio threshold {0.30…0.60} × majority floor {0.20…0.50} — proves the **40% majority floor** is the binding constraint, NOT the body-ratio threshold. At production floor 0.40 the classifier fires in **1 of 800 sessions**. Lowering the floor to **0.25** raises the fire rate to **9.75%** with **~90% directional accuracy** (vs 66% base-rate baseline). Read-only — recommendation needs Will's nod.

Headline numbers

- Sample: **800 (symbol, session) pairs**, 387 unique symbols, 9 trading dates (2026-04-09 → 2026-04-19). 45 437 5-min RTH bars total.
- Bar-level body-ratio distribution: median **0.558**, IQR [0.388, 0.730]. **55% of real bars** already exceed body > 0.50 — so 'strong-body' is common.
- At production thresholds (body=0.50, floor=0.40), classifier fires in **1 of 800** sessions (0.12%). On that single fire, the vote was **wrong**.
- Lowering the floor to **0.25**: fire rate **9.75%** (78 sessions), accuracy **~90%**.
- Decomposed by side: up-pred **92.2%** (vs 66% base), down-pred **87.0%** (vs 34% base). Both sides beat their base rate by **+20 to +25 pp**.
- **Body-ratio threshold sweep is not the answer.** Even at body=0.30, fire rate is 1.9%. The floor was the lever all along.

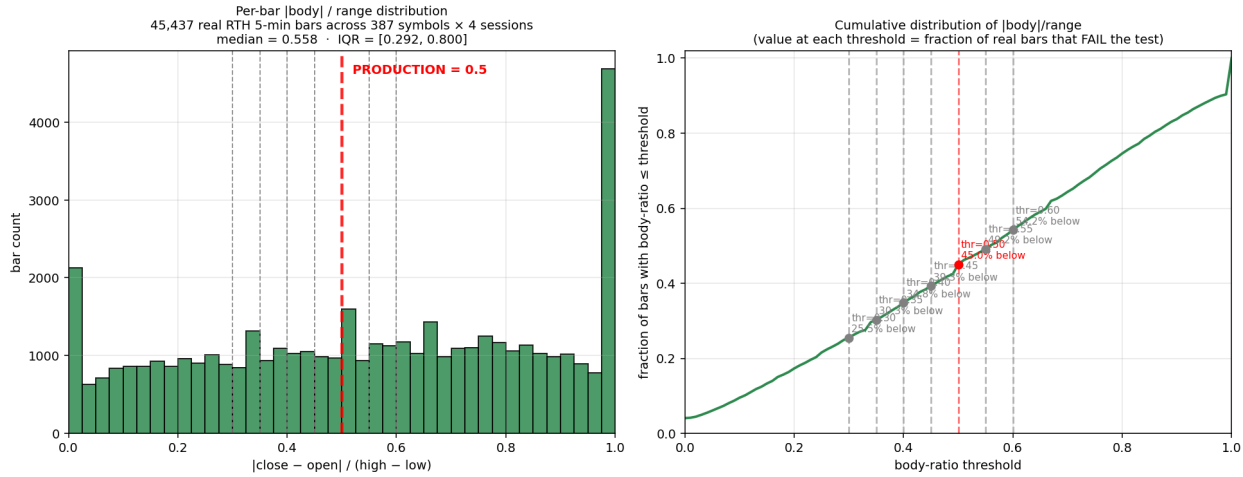
The headline figure — floor sweep



Left: per-floor session fire rate (red bar = production 0.40). Right: directional accuracy of the classifier's vote against realised open → close session move. Production = 1 fire in 800 (0% accuracy on that one fire). Floor 0.25 = 78 fires, 90.5% accuracy.

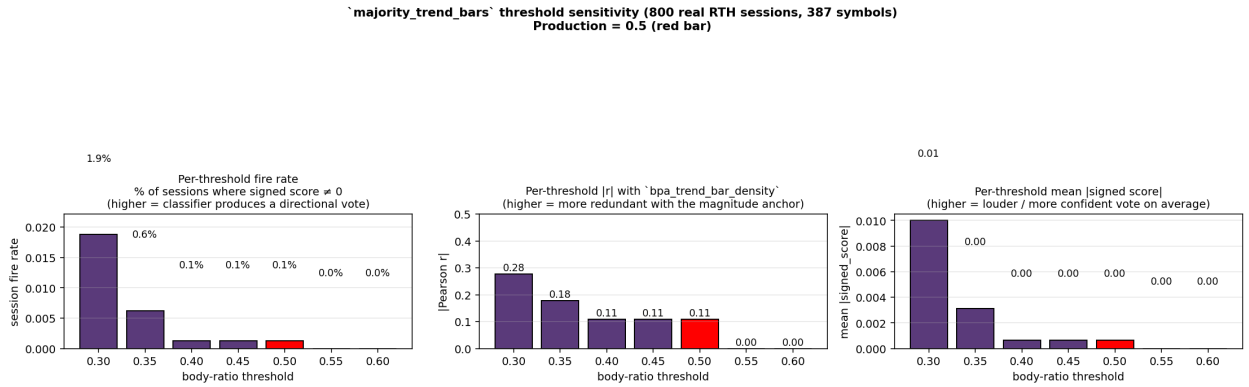
Why the production floor produces dead silence

Per-bar instrumentation explains it cleanly. ~50% of bars are bull, ~50% bear. ~55% have body > 0.50. Intersection (strong-body bars in *one direction*) ≈ **25%** of bars per session. Production floor of 40% sits well above that. The 1/800 fire rate is exactly what the math predicts.



Left: per-bar |body|/range histogram across 45 437 real RTH bars. Median 0.558. Right: cumulative distribution showing what fraction of bars are FILTERED OUT at each candidate body-ratio threshold.

Body-ratio sweep — confirms the body threshold isn't the lever



Three panels: per-threshold session fire rate, |r| with bpa_trend_bar_density, mean |signed score|. All three drop near zero across the range. The body threshold isn't the binding constraint — the majority-floor 40% gate is.

What's next — needs Will's nod

- **RECOMMENDED:** introduce **MAJORITY_TREND_BAR_FLOOR = 0.25** in aiedge/signals/components.py and have **_score_majority_trend_bars** use it. Body-ratio threshold stays at 0.50.
- Re-baseline replay-equivalence tests (**MajorityTrendBarsReplayEquivalence**) under the new floor.
- Re-run the incr-17 redundancy study after the change — current |r|_avg = 0.158 will shift; need to remeasure correlation with bpa_trend_bar_density (the magnitude-family anchor).
- Pattern Lab WR study still blocked on DB backfill.
- Larger sample. 800 sessions across 9 trading dates is enough to surface this; multi-month would let the same study run on overnight ES too.

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification